SAI SATWIK BIKUMANDLA

Software Engineer | Full Stack Developer | AI Engineer (LLM Applications)

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PROFESSIONAL SUMMARY

Software Engineer with an M.S. in Computer Science (University at Albany, May 2026) who builds full-stack web applications, RESTful APIs and AI agent systems with Python, Java, TypeScript, React, Next.js, FastAPI and SQL. Designed, deployed and operate a self-hosted, LLM-powered AI agent (Ollama, Qwen3) that processes ~13,000 job postings per night on Linux, with human-in-the-loop approval, least-privilege security and continuous deployment with automatic rollback. Works with MySQL, MongoDB, AWS, GCP, Docker and Kubernetes. First author of a peer-reviewed deep learning paper.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, C++, PHP, HTML5, CSS3, Bash
Frontend: React.js, Next.js, Tailwind CSS, Responsive Design, Accessibility (WCAG), Figma, UI/UX Design
Backend & APIs: FastAPI, Flask, Node.js, RESTful APIs, Microservices, JSON, JWT, NextAuth, Zod, Spring Boot (in progress)
AI & Machine Learning: Large Language Models (LLMs), Generative AI, Agentic AI, AI Agents, Tool Calling, Prompt Engineering, Structured Outputs (JSON Schema), Human-in-the-Loop Guardrails, Model Evaluation, Ollama, Qwen3, Claude API (Anthropic), Gemini API, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, NLP, Word Embeddings, Computer Vision, CNN, LSTM
Databases: MySQL, MongoDB, Google Cloud SQL, Relational Database Design, Schema Design, Query Optimization, Indexing
Cloud & DevOps: AWS, Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Git, GitHub, Linux, systemd, Vercel
Practices: Object-Oriented Programming (OOP), Data Structures & Algorithms, System Design, API Design, Software Development Life Cycle (SDLC), Agile/Scrum, Unit Testing, Debugging, Troubleshooting, Code Review, Problem-Solving

PROFESSIONAL EXPERIENCE

Software Development Intern | EcoServants (Ecological Servants Project) | Remote **Oct 2026 – Present**

- Develop the Final-Email Preflight Validator for GreenSend, an end-to-end assignment on the tech development team.
- Report progress and blockers to the engineering supervisor in weekly tech development meetings.

Technical Operations & Software Support Associate | Real Craft Tech | Remote **Mar 2024 – Jun 2024**

- Tracked 15+ concurrent technical and operational work items for a remote team, following blockers through resolution.
- Translated leadership decisions and business requirements into 40+ actionable technical tasks with clear owners.
- Managed 50+ recurring stakeholder requests, keeping requesters, decision makers and implementers informed at each stage.
- Documented 30+ processes, decisions and operational records in a shared knowledge base to speed up team handoffs.
- Organized cross-functional workflows and standardized task-tracking templates across 10+ concurrent priorities.
- Prepared 20+ status reports on progress, dependencies and open issues for technical and non-technical stakeholders.

Technical Customer Operations & Systems Support Executive | Real Craft Tech | Remote **Mar 2023 – Jul 2023**

- Provided technical support in 200+ customer interactions, diagnosing issues and writing them up as structured requirements.
- Tracked 80+ open customer issues through resolution and monitored response times to meet service expectations.
- Routed 150+ requests to the right technical or operational owner as the liaison between customers and internal teams.
- Documented requirements, issues and follow-up actions across 100+ cases, keeping 120+ ongoing requests traceable.
- Analyzed recurring customer requests and feedback to identify 15+ common issues and process-improvement opportunities.
- Gathered requirements in direct customer conversations and converted 75+ requests into actionable tasks.

Web Development Intern | Rethink UX | Remote **Oct 2022 – Dec 2022**

- Developed 12+ client-facing website features with PHP, WordPress, JavaScript, HTML5 and CSS3, from requirements to release.
- Implemented role-based access control (RBAC) across 3+ user roles to govern feature visibility and application behavior.
- Built frontend UI logic for 20+ responsive website components, turning functional requirements into maintainable code.
- Debugged and resolved 25+ frontend and access-control defects, improving application reliability.
- Integrated 10+ features into existing application workflows, coordinating scope and dependencies with the development team.
- Participated in 15+ code reviews with senior engineers and wrote technical documentation for 12+ delivered features.

Programming & Technical Content Intern | Study Experts | Remote **Jun 2022 – Aug 2022**

- Authored 25+ programming tutorials on Python, Java, data structures and algorithms for technical learners.
- Developed 50+ runnable Python and Java code examples demonstrating OOP, data structures and algorithms.
- Tested and debugged every code example before publication to verify correctness and execution behavior.
- Reviewed 70+ technical articles for correctness, code quality, clarity and consistency.
- Explained 30+ computer science topics in step-by-step language for beginners.
- Maintained and revised knowledge-base resources in a collaborative content workflow.

PROJECTS

Job Agent: Self-Hosted LLM AI Agent | *Python*, *FastAPI*, *Ollama*, *Qwen3*, *JavaScript*, *Linux*, *systemd* | GitHub **2026**

- Built an LLM-powered AI agent that ingests ~13,000 job postings nightly from ~140 career boards via the Greenhouse, Lever, Ashby, Workday and SmartRecruiters REST APIs, using only local open-source LLMs.
- Used structured outputs (JSON Schema) so a Qwen3 4B model extracts required skills while Python computes deterministic match scores; a Qwen3 8B model drafts tailored resumes and cover letters for top matches.
- Reduced prompt processing per agent step from ~30 s to 1.7 s (about 17x) on a CPU-only Intel Core i3 server by benchmarking inference and redesigning prompts around Ollama's prompt cache.
- Built a tool-calling agent loop (shell, file, web fetch and web search tools) with human-in-the-loop approval for every side-effecting action, served through a FastAPI web panel with ETag caching.
- Secured the system with least-privilege Unix users, one scoped sudo rule and UID-matched firewall rules, and exposed the FastAPI web panel only over a private Tailscale network (VPN) to authorized devices.
- Automated continuous deployment (CI/CD) with a systemd timer that pulls from GitHub, runs syntax checks, backs up the running release and rolls back automatically on a failed health check.
- Wrote a public incident postmortem tracing a 95°C thermal alert to 32-second CPU turbo bursts using systemd journal and sar data, then shipped a 25 W power cap, heat guard and alert logging.

LaTeX Resume Builder: Full-Stack SaaS Web Application | *Next.js*, *React*, *TypeScript*, *MongoDB*, *NextAuth*, *Docker* | GitHub **2025**

- Built a full-stack web application with Git-style resume repositories, commit history and rollback to any prior version.
- Designed RESTful APIs for repositories, versions and authentication on an append-only MongoDB data model.
- Built a dual-rendering pipeline: a live HTML preview plus a server-side LaTeX compilation API that exports PDFs.
- Protected routes with NextAuth, validated input with Zod, rate-limited login, and containerized the app with Docker.

MyDishDB: Cloud Relational Database | *SQL*, *MySQL*, *Google Cloud SQL*, *JWT* **2025**

- Designed a normalized relational schema of 10+ entities with primary and foreign keys, CHECK constraints and triggers.
- Deployed on Google Cloud SQL with role-based access control and a JWT backend; optimized multi-table JOINs with indexes.

Tennis Ball Detection & Tracking: Computer Vision Pipeline | *Python*, *Roboflow API*, *SORT*, *Gradio* **2025**

- Integrated the Roboflow object-detection REST API into two tracking pipelines with anti-jump filtering, exponential smoothing, velocity extrapolation and SORT multi-object tracking.
- Built a TrackingAnalytics engine that exports per-session JSON data to an interactive Gradio web UI with session history.

Web Prototype Generator: AI Code Generation | *Python*, *Flask*, *Gemini API*, *Prompt Engineering* | GitHub **2024**

- Led the Gemini API integration in a Flask backend that turns natural-language briefs into responsive HTML/CSS/JS prototypes.
- Built the prompt-engineering pipeline and set AI-attribution tagging standards the team adopted.

Movie Recommendation System: NLP Recommender | *Python*, *scikit-learn*, *TF-IDF*, *NLP* | GitHub **2025**

- Built a content-based recommender over 5,000+ films using TF-IDF vectorization, cosine similarity and NLP preprocessing.
- Evaluated relevance with Precision@K at K = 5, 10 and 20 and tuned similarity thresholds against a random baseline.

Additional Projects | *Python*, *Streamlit*, *JavaScript* **2025–2026**

- Web Security Threat Scanner: Python tool that scores website risk across 13 weighted security categories.
- MarkVid: video-bookmarking Chrome extension built with JavaScript and published on the Chrome Web Store.

EDUCATION

Master of Science (M.S.) in Computer Science | University at Albany, State University of New York | Albany, NY **May 2026**
Coursework: Operating Systems, Database Systems, Computer Security, Algorithms & Data Structures, AI, Computer Vision

B.Tech., Computer Science & Information Technology | Sri Indu College of Engineering & Technology (JNTUH) | India **Apr 2024**
Coursework: Machine Learning, Artificial Intelligence, Software Engineering, Software Testing Methodologies, Computer Networks

PUBLICATIONS

First author, "Program for Emotion-Responsive MIDI Proposal," International Journal of Engineering and Techniques (IJET), Vol. 11, Issue 3, May 2024. CNN facial-emotion classifier with 95.14% accuracy on FER2013 (30,219 images).

CERTIFICATIONS

Building AI Agents: Advanced Techniques for Developers, LinkedIn Learning (Aug 2026) | Google AI Essentials, Google (Jul 2026) | Building with the Claude API, Anthropic (Mar 2026)